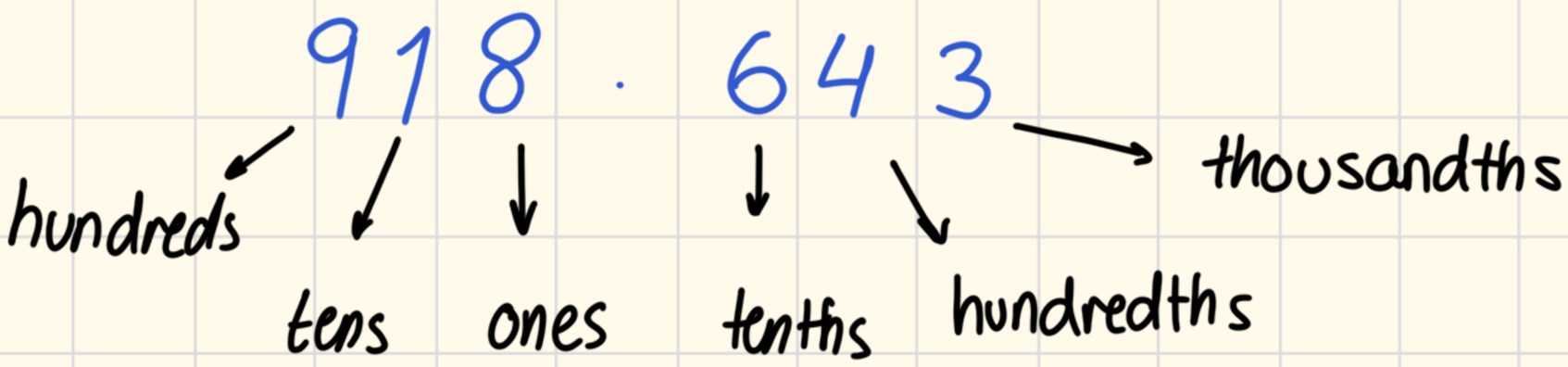
Approximations And Bounds

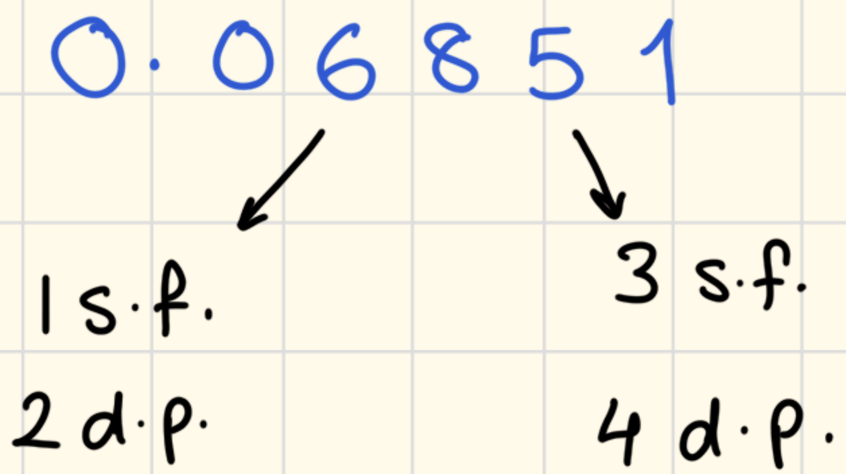
To approximate a degree of accuracy,

1. Underline Degree
2. Look At Digit After
3. < 5 Keep Degree | ≥ 5 Degree + 1

Degrees Of accuracy

- tenth $\frac{1}{10}$
- hundredths $\frac{1}{100}$





Approximations And Bounds

Approximate 10.963 to the nearest

a)
- tens — 10.963 — 10
- ones — 10.963 — 11
- tenths — 10.963 — 11.0

b)
- 1 d.p. — 10.963 — 11.0
- 2 d.p. — 10.963 — 11.96

c)
- 2 s.f. — 10.963 — 11
- 3 s.f. — 10.963 — 11.0

Approximations And Bounds

To get bounds with degree of accuracy,

1. Underline Degree
2. Upper Bound: Put 5 After Degree
3. Lower Bound: Put 5 After Degree, Degree - 1

Approximations And Bounds

Find bounds 10.963 to the nearest 8

		Upper Bound	Lower Bound
a) • tens	10	15	5
• ones	10	10.5	9.5
• tenths	10.9	10.95	10.85
b) • 1 d.p.	10.9	10.95	10.85
• 2 d.p.	10.96	10.965	10.955
c) • 2 s.f.	10	10.5	9.5
• 3 s.f.	10.9	10.95	10.85